MIDWEST GEOLOGICAL SEQUESTRATION CONSORTIUM

FAQs

Frequently Asked Questions

Is carbon sequestration safe?

Yes, if the storage site is correctly selected, designed, and operated. Extensive monitoring takes place during and after injection to be sure the CO_2 being stored stays in place. Monitoring techniques include using geophysical technology to confirm the position of the CO_2 underground and wells to monitor groundwater and soils. For decades, the ISGS has collected background information on the subsurface rocks of the Illinois Basin and has been investigating the sequestration potential of the Mt. Simon Sandstone for more than 10 years. The uppermost Mt. Simon Sandstone has performed well as a natural gas storage reservoir in Illinois, and we expect the same performance for the deep Mt. Simon in Decatur and other locations throughout Illinois, Kentucky, and Indiana.

How much CO_2 can the Mt. Simon Sandstone hold?

The Mt. Simon Sandstone is commonly used for natural gas storage in the Illinois Basin and has fair to good permeability and porosity. Its overlying strata contain impermeable limestone, dolomite, and shale intervals. The depth of the Mt. Simon ranges from approximately 2,000 to 14,000 ft below the surface. In the southern half of the Basin the reservoir is brine-filled, and no oil or natural gas resources have been discovered in this unit. At its greatest thickness in the Illinois Basin, the Mt. Simon is over 2,600 ft thick. The estimated storage capacity of the Mt. Simon Sandstone ranges from 11 to 150 billion tonnes of CO_2. Given the current total emissions of 291 million tonnes per year from point sources in the Illinois Basin, the Mt. Simon Sandstone has several hundred years of storage potential.







Are there any risks to pumping so much CO_2 underground?

No significant health, safety, or environmental risks should arise from a properly designed and managed storage site. Appropriate risk mitigation and management plans are an integral part of the overall project planning. Extensive monitoring takes place during and after the injection to be sure the CO_2 stays in place. The first line of monitoring begins deep below the ground, so we will know if any leakages occur long before the CO_2 reaches the surface.

How many other projects like this exist around the world?

Geological sequestration of CO_2 by injection into the Earth's subsurface is a promising technology being studied around the world. There are currently projects underway in Canada, Norway, Algeria, Australia, and under development around the world. The Illinois Basin – Decatur project is one of the first projects of similar scope in the United States to begin injection of CO_2 for geologic sequestration.